

Jiankun(Jake) Zhang

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EDUCATION

University of Chicago — Chicago, IL

B.S. in Computer Science & B.A. in Economics | Expected Jun 2028 | GPA: 3.7/4.0

- Coursework: Machine Learning, Algorithms, Data Structures, System Programming, Computer Graphics, Discrete Math, Macroecon, Statistics.
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EXPERIENCE

MIT CSAIL — Cambridge, MA

Research Assistant | Mentored by Zhiyang (Frank) Dou | Jun 2026 – Present

- Built benchmark infrastructure for Robot Skill Puzzles (RSP), a decentralized skill-prior framework for humanoid control, integrating InterMimic/SONIC environments in NVIDIA Isaac Lab and scaling training from 256 to 1,024 parallel environments.
- Developed Python/Bash infrastructure for checkpointing, evaluation, multi-seed aggregation, GPU resource management, and automated performance, memory, and cost analysis.
- Implemented parameter-matching, refrence-motion validation, and reproducible evaluation pipelines for humanoid RL experiments.

Agile Robots — Munich, Germany

Humanoid Reinforcement Learning Intern | Jun 2026 – Aug 2026

- Optimized a GPU-accelerated humanoid RL pipeline by replacing per-environment Python/PhysX queries with batched GPU tensor operations, increasing simulation throughput from ~275 to 25,000+ steps/s ($>100\times$).
- Extended distributed PPO training in PyTorch across 2 GPUs with gradient accumulation and cross-rank synchronization, supporting 196K-sample effective batches and validated by 42 unit and regression tests.
- Scaled Unitree G1 reinforcement learning to 4 GPUs and 32,768 parallel simulation environments.

Ohio State University — AIoT and Machine Learning Lab — Columbus, OH

Research Assistant | Mar 2025 – Present

- Co-authored RespiraMFM (ACL 2026); built data pipelines to extract, normalize, and convert heterogeneous patient metadata across 7 real-world respiratory datasets into standardized natural-language inputs for multimodal LLM training, contributing to +9.15% supervised and +20.98% zero-shot AUROC across 9 disease-classification tasks.
 - Constructed LLM-agent benchmark tasks for SkillEvolBench by generating and systematically perturbing procedural workflows with information gaps, context shifts, adversarial shortcuts, and failure conditions to evaluate skill transfer and composition.
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PROJECTS

AgentMeter | [Open VSX](#) | [GitHub](#)

TypeScript/JavaScript · VS Code Extension API · GitHub Actions · CI/CD | 2026

- Built and published a VS Code/Cursor extension for monitoring AI coding-tool usage and quotas, reaching 1,300+ downloads on Open VSX.
- Developed editor-integrated usage tracking and dashboard functionality and automated extension releases through CI/CD.

Autoencoder Lab | [Website](#) | [GitHub](#)

React · TypeScript · Vite · Vitest · GitHub Actions | 2026

- Built and deployed an interactive React/TypeScript web application for visualizing AE, VAE, VQ-VAE, and FSQ architectures through latent-space and reconstruction experiments.
 - Implemented interactive Canvas/SVG visualizations, automated testing with Vitest, and CI/CD with GitHub Actions.
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PUBLICATIONS

RespiraMFM: A Multimodal Foundation Model with Contrastive Audio-Language Alignment for Respiratory Disease Identification — Co-author, ACL 2026 Main Conference | <https://arxiv.org/abs/2606.09966>

SkillEvolBench: Benchmarking the Evolution from Episodic Experience to Procedural Skills — Co-author | <https://arxiv.org/abs/2605.24117>

TECHINICAL SKILLS:

Languages: Python, Java, C, JavaScript, TypeScript, HTML/CSS

Software & Systems: Linux/Unix, Bash, Git, GitHub, PyTorch, GitHub Actions, React, Vite, Vitest

ML/AI: Reinforcement Learning (PPO), LLM Agents, Multimodal Learning, Contrastive Learning, TensorFlow, NumPy

Robotics & Simulation: NVIDIA Isaac Lab, Isaac Sim, PhysX, InterMimic, SONIC, Unitree G1, SMPL-X, OMOMO, GMR